

Hannah Liu

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Education

University of Waterloo

Bachelor of Software Engineering, Honours Co-op

Sep 2024 – Apr 2029

Waterloo, ON

Technical Skills

Languages: Python, TypeScript/JavaScript, C/C++, HTML/CSS, Ruby/Ruby on Rails, Java, SQL, GraphQL

Frameworks/Tools: React/React Native, Git, Figma, Jest, Playwright, PostgreSQL, Chromatic, Storybook, Rspec, Signadot

Work Experience

Wealthsimple

Sep 2026 - Present

Software Development Intern

Toronto, ON

- Led the end-to-end development of an insurance-enrollment system extending travel coverages to all Quebec cardholders, authoring technical briefs and managing external vendor integration with the Walnut third-party API
- Engineered the transactional insurance backend using **Ruby on Rails**, **PostgreSQL**, and **GraphQL** to process benefit enrollments and manage durable consent state across 3 card tiers for 30,000+ users
- Delivered a tier-aware credit-card de-emphasis rollout across web, mobile, marketing, and recommendation surfaces by removing and reprioritizing card promotions for certain tiers of clients, and lowering the recommendation exclusion threshold using **SQL/dbt** and TypeScript

Shopify Top-tier performance rating

Jan 2026 – Apr 2026

Software Engineering Intern

Toronto, ON

- Secured and scaled feature-access for **2+ million merchants** by engineering a **scalable feature-gating system** for Shopify Analytics; migrated legacy plan-based logic to a **centralized access control system** in **Ruby on Rails** to improve system auditability and monetization flexibility
- Empowered **2+ million merchants** with real-time business intelligence by engineering a high-scale product status filtering engine; optimized data ingestion by refining **Apache Beam ETL data pipelines** and authoring complex **ClickHouse schema migrations** to ensure high-performance analytical accuracy
- Accelerated Shopify platform performance for global users by architecting **10+ high-performance Preact web components**; reduced initial loading time by **82%** and reduced core application weight by **90%** by migrating from legacy React components while evaluating **60+** pull requests to ensure elite quality
- Modernized data accessibility for **1.15 million active merchants** by leading the end-to-end development of the Shopify Analytics data navigation system; engineered a platform-wide standard for temporal data filtering using **React/Preact** and facilitated stakeholder feedback loops to ensure a consistent, world-class user experience across the ecosystem

Shopify

May 2025 – Aug 2025

Software Engineering Intern

Toronto, ON

- Increased test reliability from **30% to 99%** by writing and maintaining **end-to-end, unit and integration** tests using **Jest** and **Playwright**, monitoring for failures using an internal **Prometheus and Grafana** platform
- Engineered interactive UI features for the Shopify Analytics Dashboard used by **2+ million** Shopify merchants, optimizing consumer insights through **complex state management** for features like draggable metrics and animated transitions

Projects

DoorSense | React Native, Expo, TypeScript, CSS



- Designed and developed a smart home security mobile **iOS** application using **React Native**, **Expo**, and **TypeScript**
- Built responsive UI components to display live video streams and security notifications, coordinating with backend developers to seamlessly connect the mobile client

ShopSmart | React Native, TypeScript, CSS, Expo



- Led the process of designing and developing ShopSmart, an **iOS** mobile app that generates optimized grocery shopping routes based on a list of items, factoring in prices, travel time, and store proximity
- Developed responsive UI components to dynamically render optimized shopping paths, integrating with backend routing services to display real-time prices, travel times, and store proximity

Undead Uprising | Python/Pygame, OOP, Figma



- Led a team of six in developing a 2D survival game using **Python** and **Pygame**, leveraging object-oriented programming (OOP) principles to manage game states, collision detection, and sprite spawning